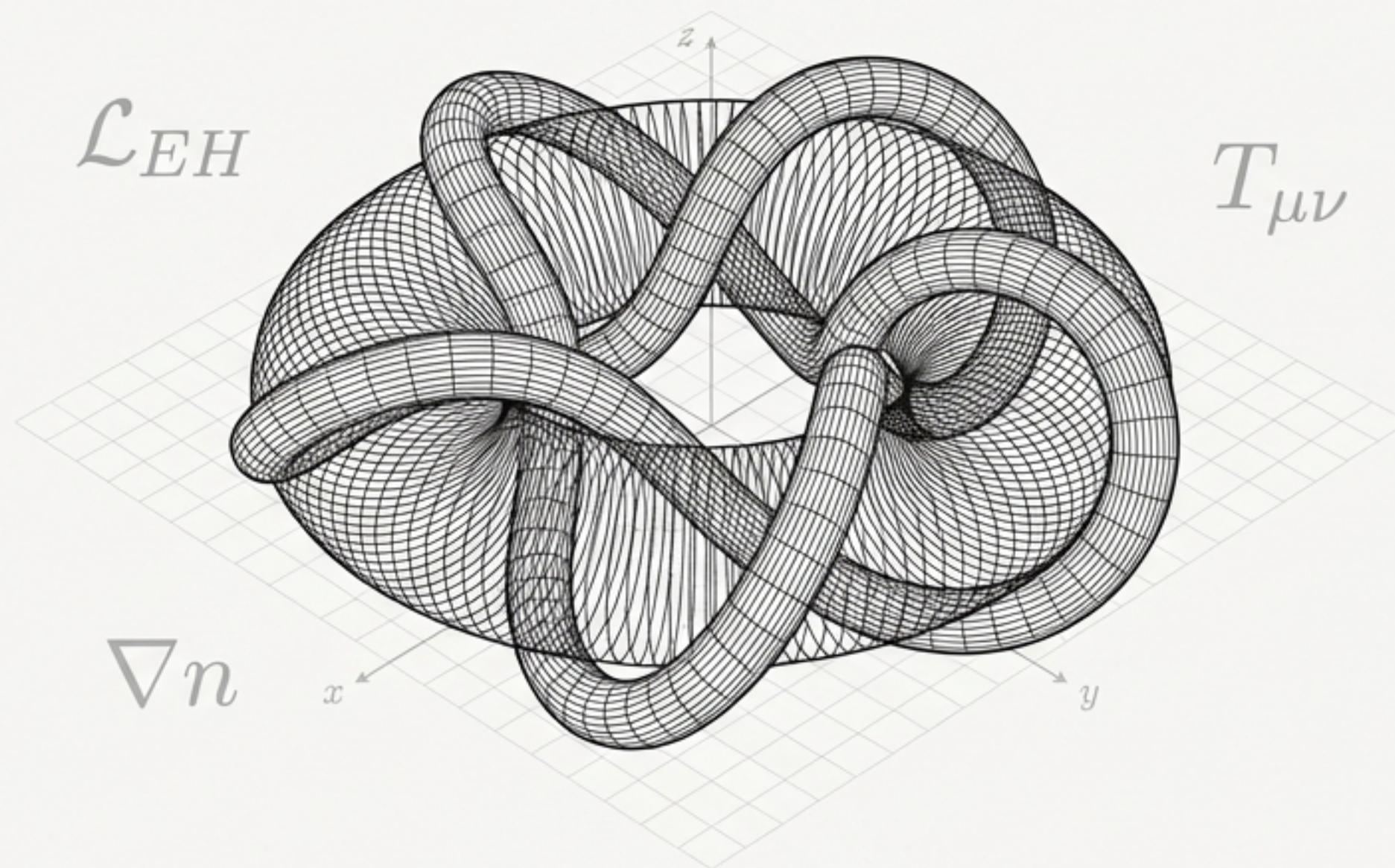


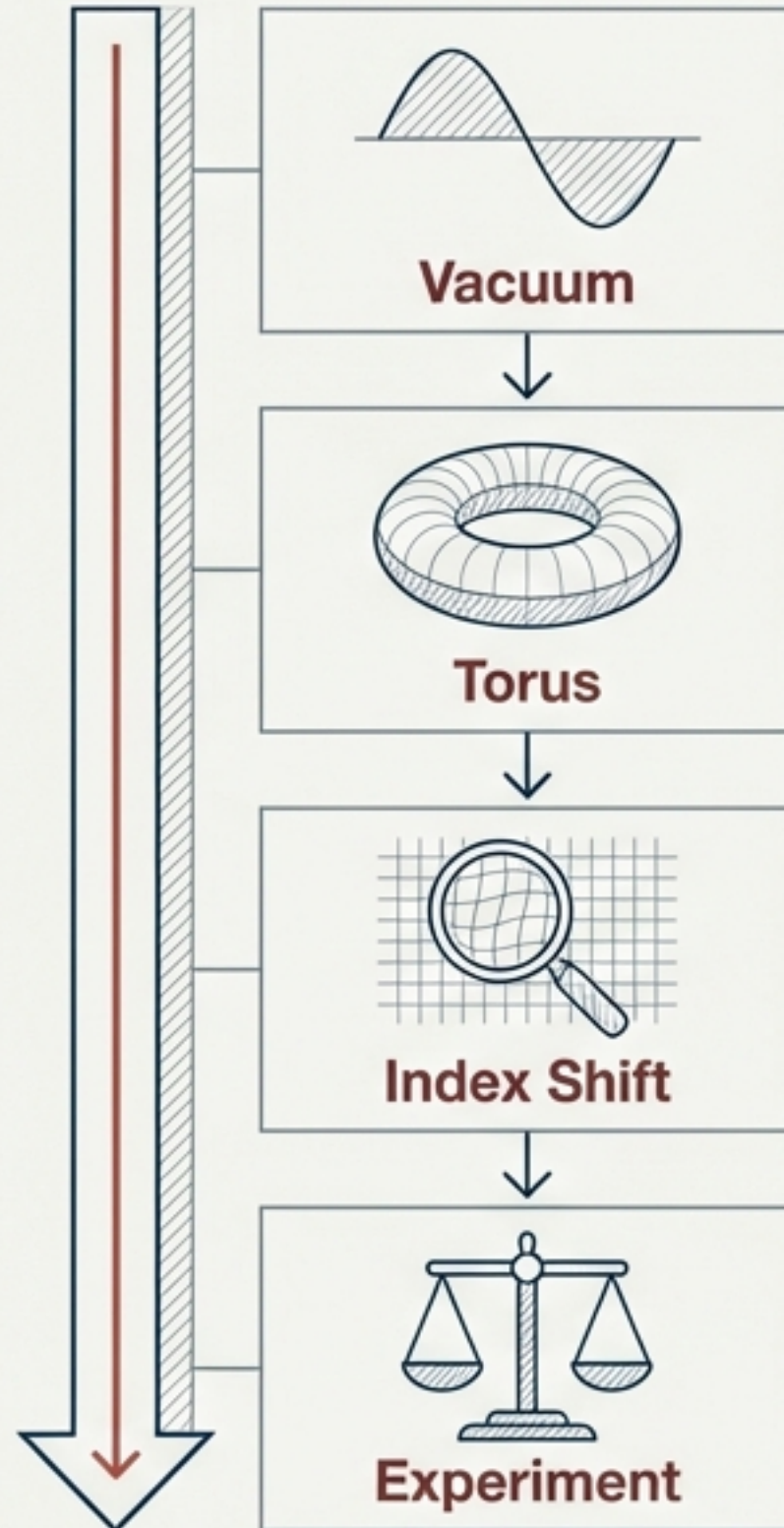
Vacuum Stress–Energy Engineering via Dynamic Toroidal Multipoles

A Framework for Laboratory-Scale Analog Gravity Experiments



Executive Summary

The Rigorous Frontier



1. Premise

The Quantum Vacuum is not empty. It is a nonlinear medium described by the Euler-Heisenberg Lagrangian, susceptible to polarization by extreme fields.

2. Innovation

Toroidal Anapole fields allow for the concentration of electromagnetic stress-energy in the near field without radiative losses, solving the containment problem.

3. Mechanism

This localized vacuum polarization creates a refractive index shift. In the Polarizable Vacuum (PV) formalism, this is mathematically equivalent to a weak metric perturbation.

4. Proposal

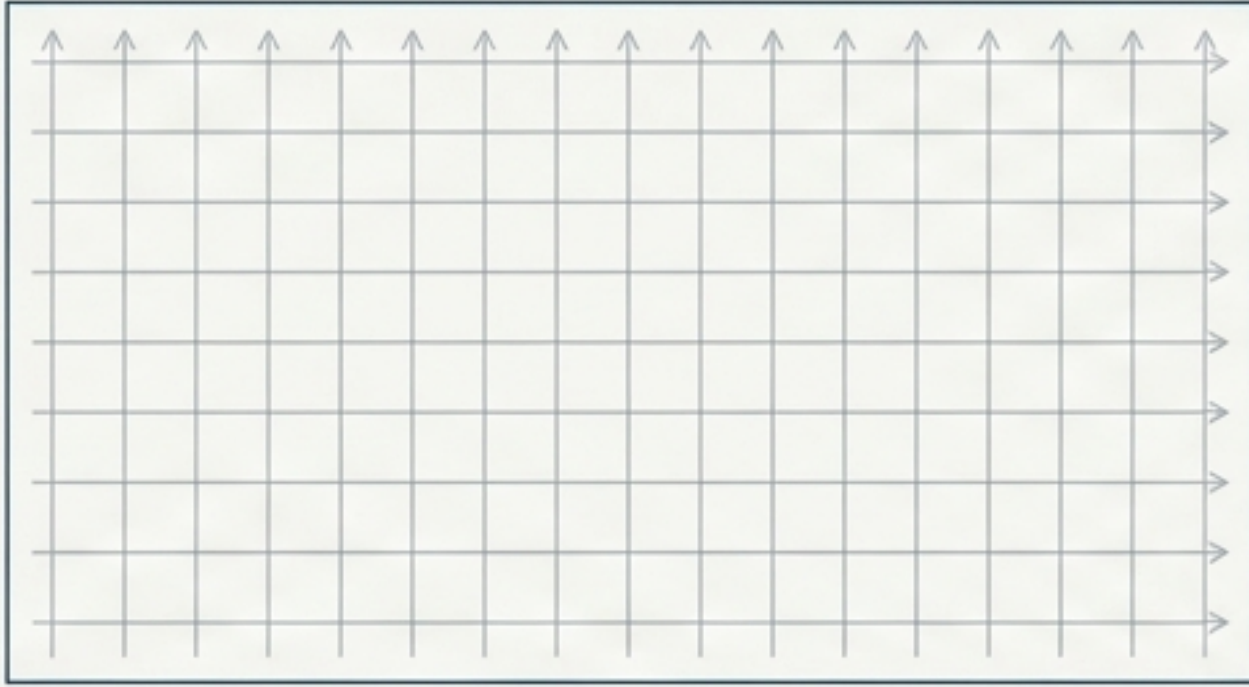
We propose a laboratory-scale null-test experiment to detect this coupling using differential measurement channels.

Key Metric

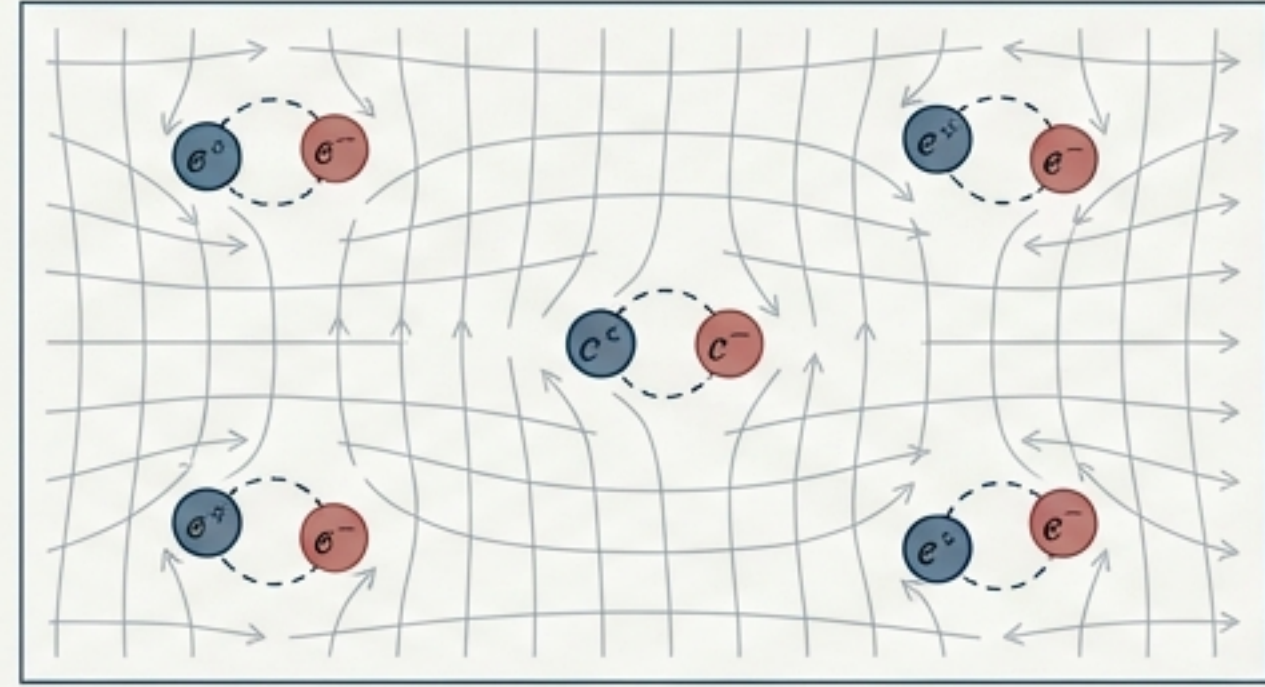
PREDICTED EFFECT MAGNITUDE: $\sim 10^{-18} \text{ m/s}^2$

Target: Precision Null-Test, not macroscopic propulsion.

The Vacuum is Not Empty: Nonlinear QED Response



Classical Vacuum
Linear Medium (ϵ_0, μ_0 constant)



QED Vacuum
Nonlinear Medium (Field-dependent ϵ, μ)

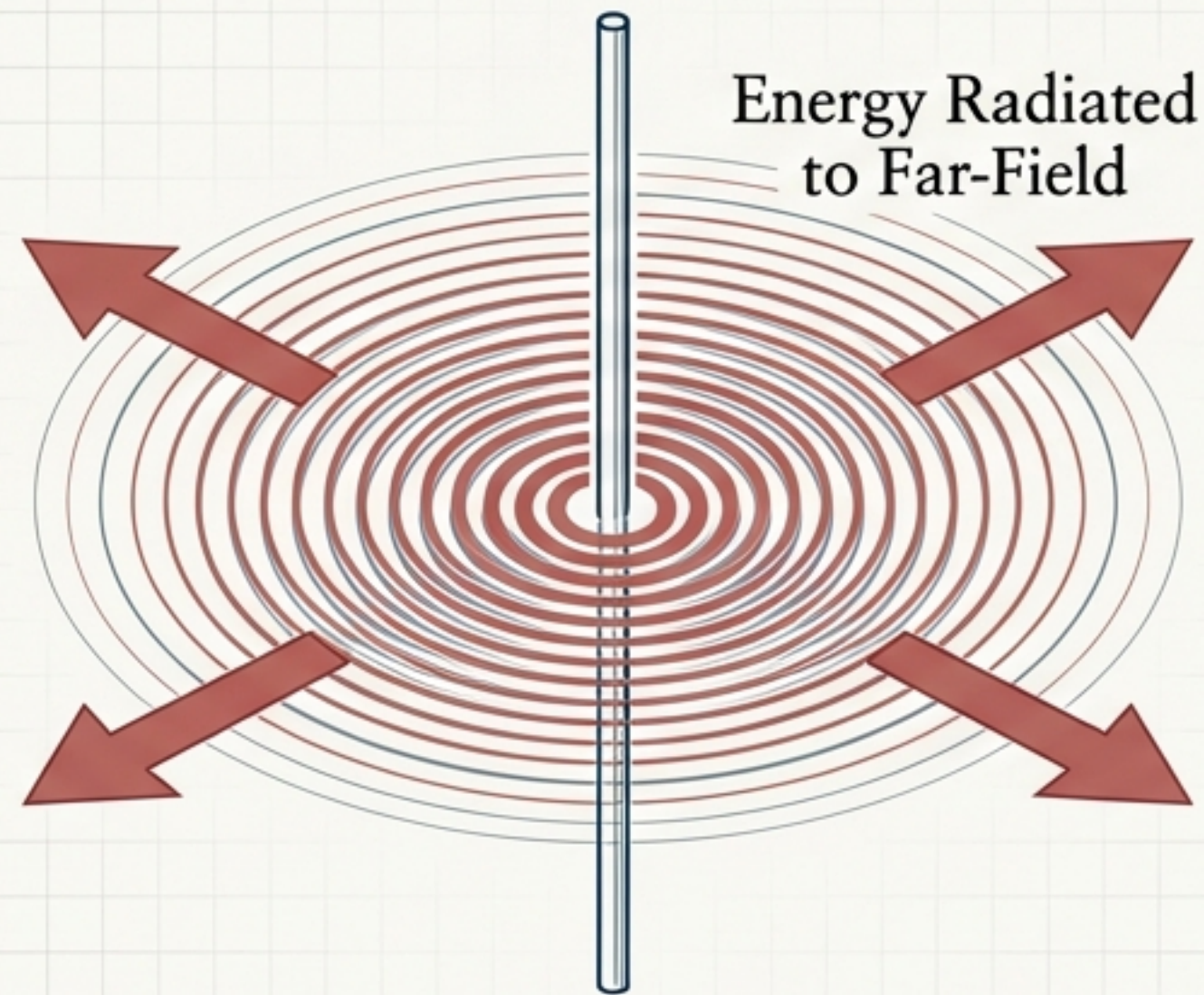
In QED, virtual fluctuations endow the vacuum with nonlinear properties. This is described by the Euler-Heisenberg Effective Lagrangian:

$$\mathcal{L}_{\text{EH}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{2\alpha^2}{45m_e^4} \left[(F_{\mu\nu}F^{\mu\nu})^2 + \frac{7}{4}(F_{\mu\nu}\tilde{F}^{\mu\nu})^2 \right]$$

Depends on Lorentz Invariants: $2(B^2 - E^2)$ and $-4(E \cdot B)$

The Engineering Challenge: Concentration Without Radiation

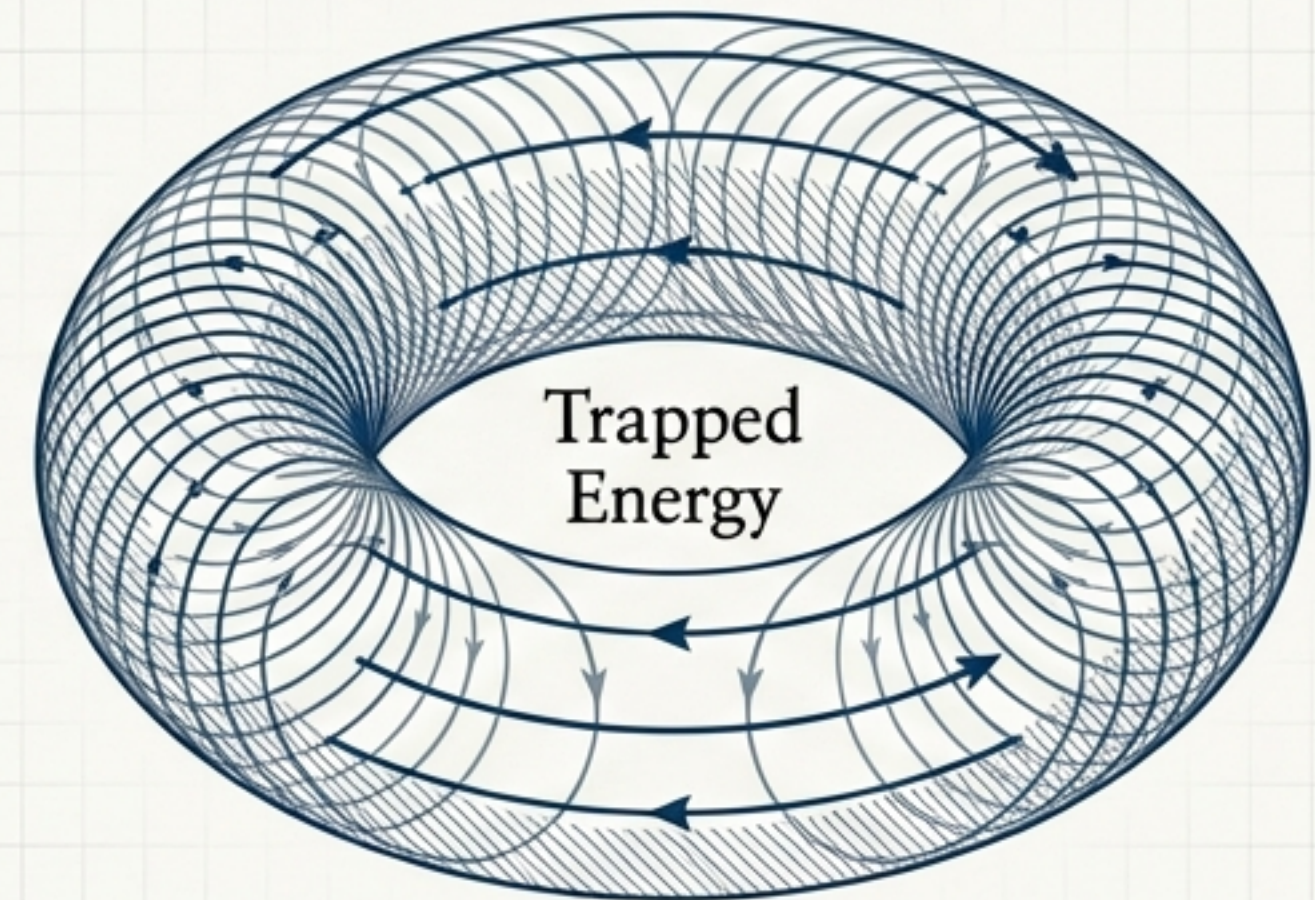
The Problem: Standard Dipole



Oscillating dipoles radiate energy efficiently. Impossible to maintain high near-field stress without massive power loss.

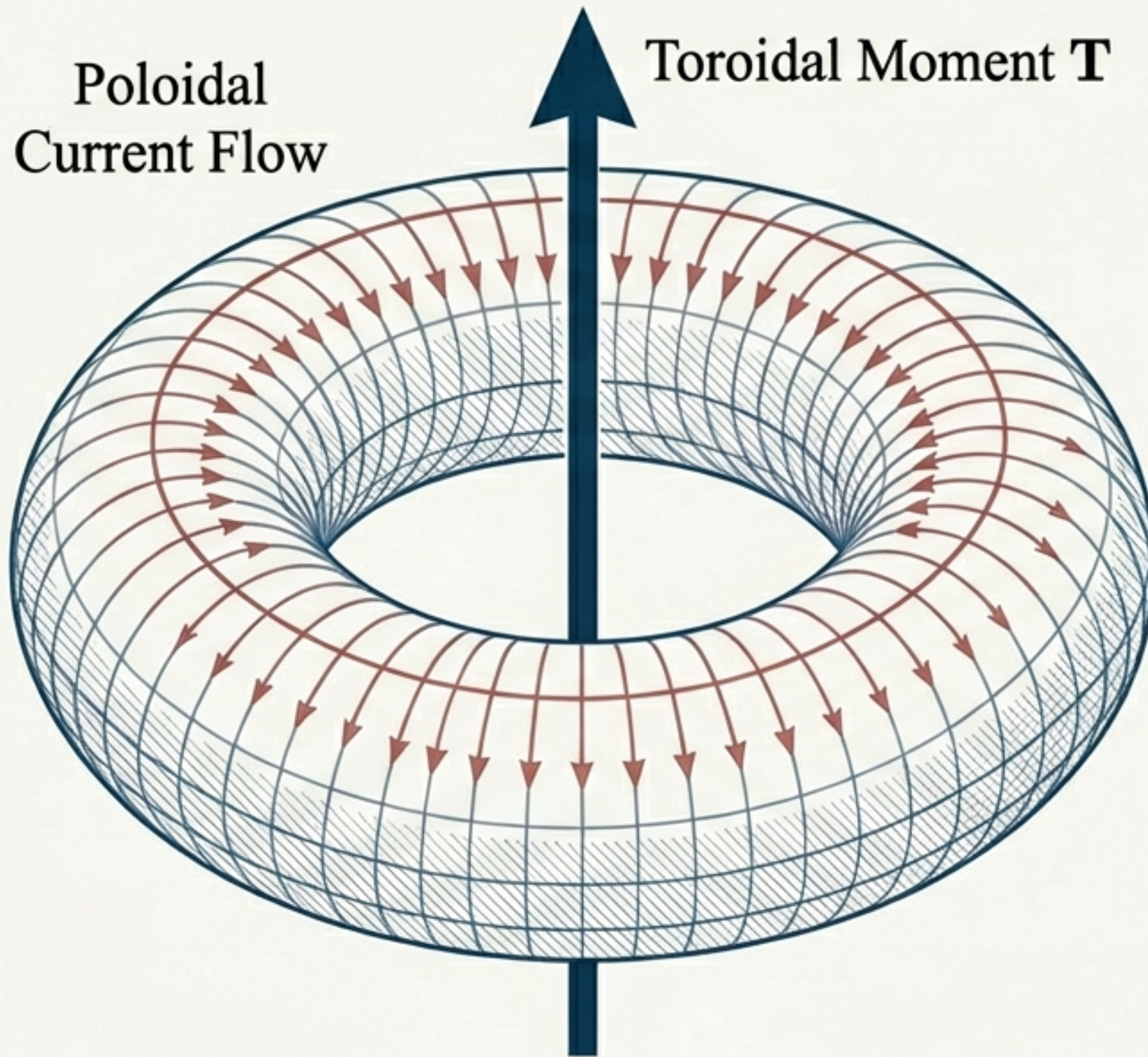
Constraint: We must maximize $E^2 - B^2$ and $E \cdot B$ locally while minimizing radiation.

The Solution: Toroidal Container



Toroidal topology minimizes far-field emission. Energy is “contained” in the near-field, maximizing stress invariants.

Beyond the Dipole: The Toroidal Moment



Distinct Topology: Unlike electric or magnetic dipoles, the Toroidal moment corresponds to poloidal currents.

Equation 5: Definition of Toroidal Moment

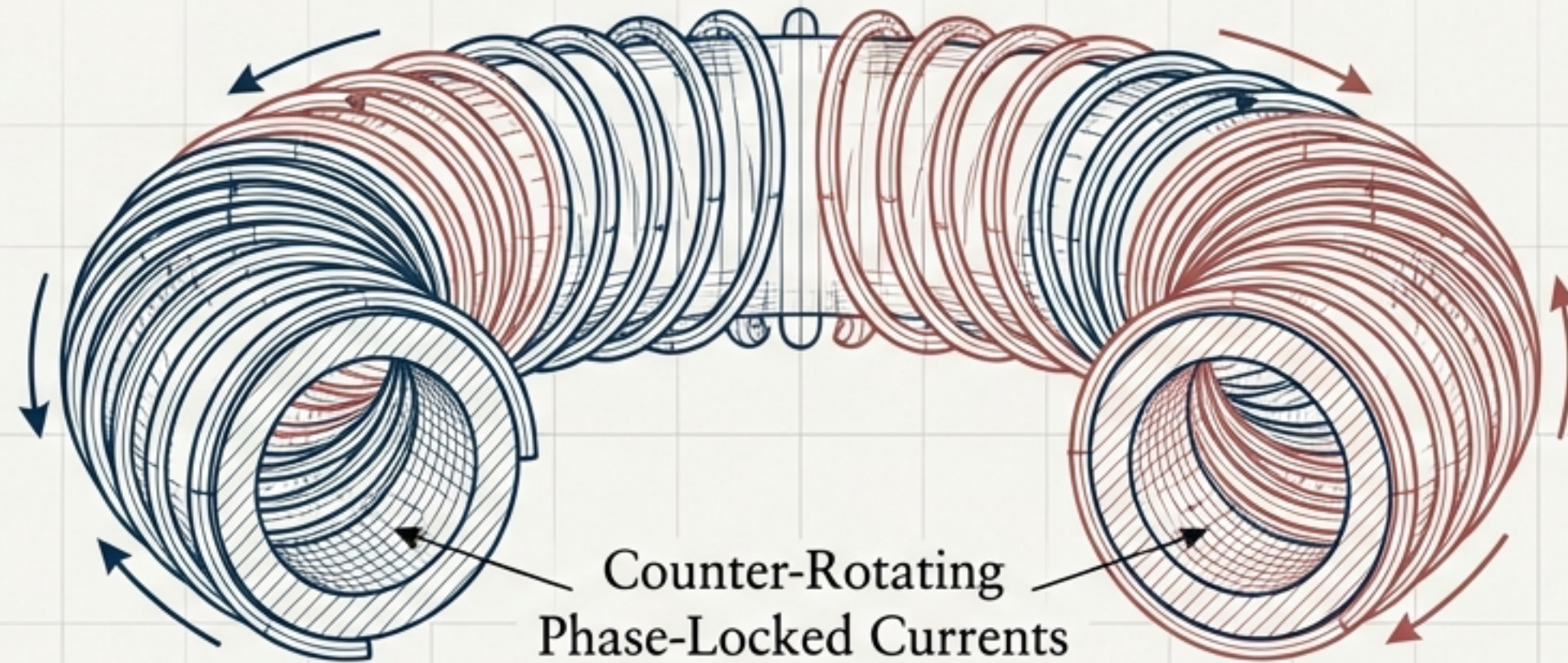
$$\mathbf{T} = \frac{1}{10c} \int [(\mathbf{r} \cdot \mathbf{J})\mathbf{r} - 2r^2\mathbf{J}] d^3r$$

Key Properties

- Associated with finite magnetic helicity.
- Generates longitudinal vector potentials.
- Historically elusive, but critical for non-radiating sources.

The Anapole State: Maximizing Near-Field Stress

The “Stealth” Configuration

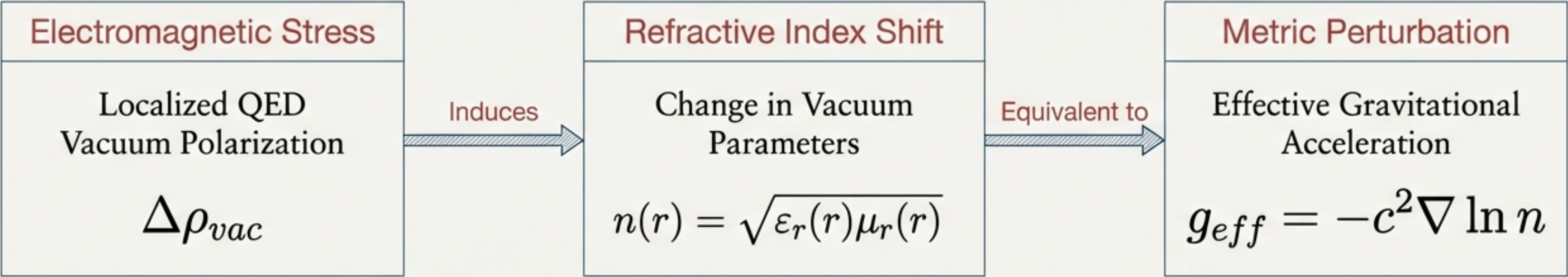


Mechanism:

1. Net Magnetic Dipole $m \approx 0$ (Fields cancel at distance)
2. Toroidal Moment T is maximized.

Radiation is suppressed through multipole cancellation. Electromagnetic stress is trapped and concentrated in the near-field region.

Bridging Fields and Gravity: The Polarizable Vacuum

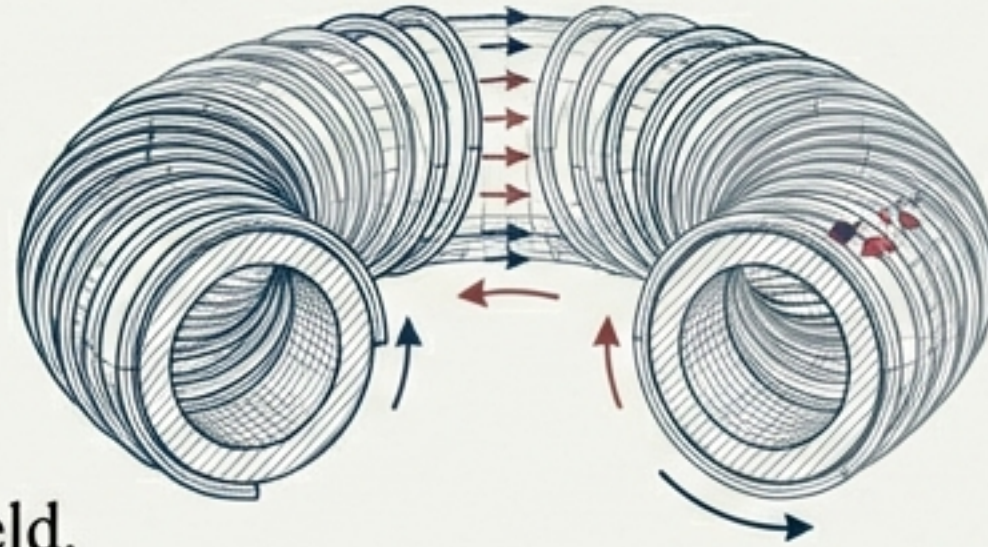


Note: This equivalence holds strictly within the Polarizable Vacuum (PV) representation of gravity.

Simulation Methodology: Maxwell–Euler–Heisenberg System

1. Classical Solution

Solve Maxwell equations for toroidal geometry using FDTD methods.



2. Compute Invariants

Calculate field invariants in the near field.

$$F_{\mu\nu}F^{\mu\nu} \text{ and } F_{\mu\nu}\tilde{F}^{\mu\nu}$$

3. EH Correction

Apply Euler-Heisenberg perturbation to find induced vacuum energy density.

$$\Delta\rho_{vac}$$

4. Index Perturbation

Reconstruct $\delta n(r)$ to derive effective metric.

Validity Check

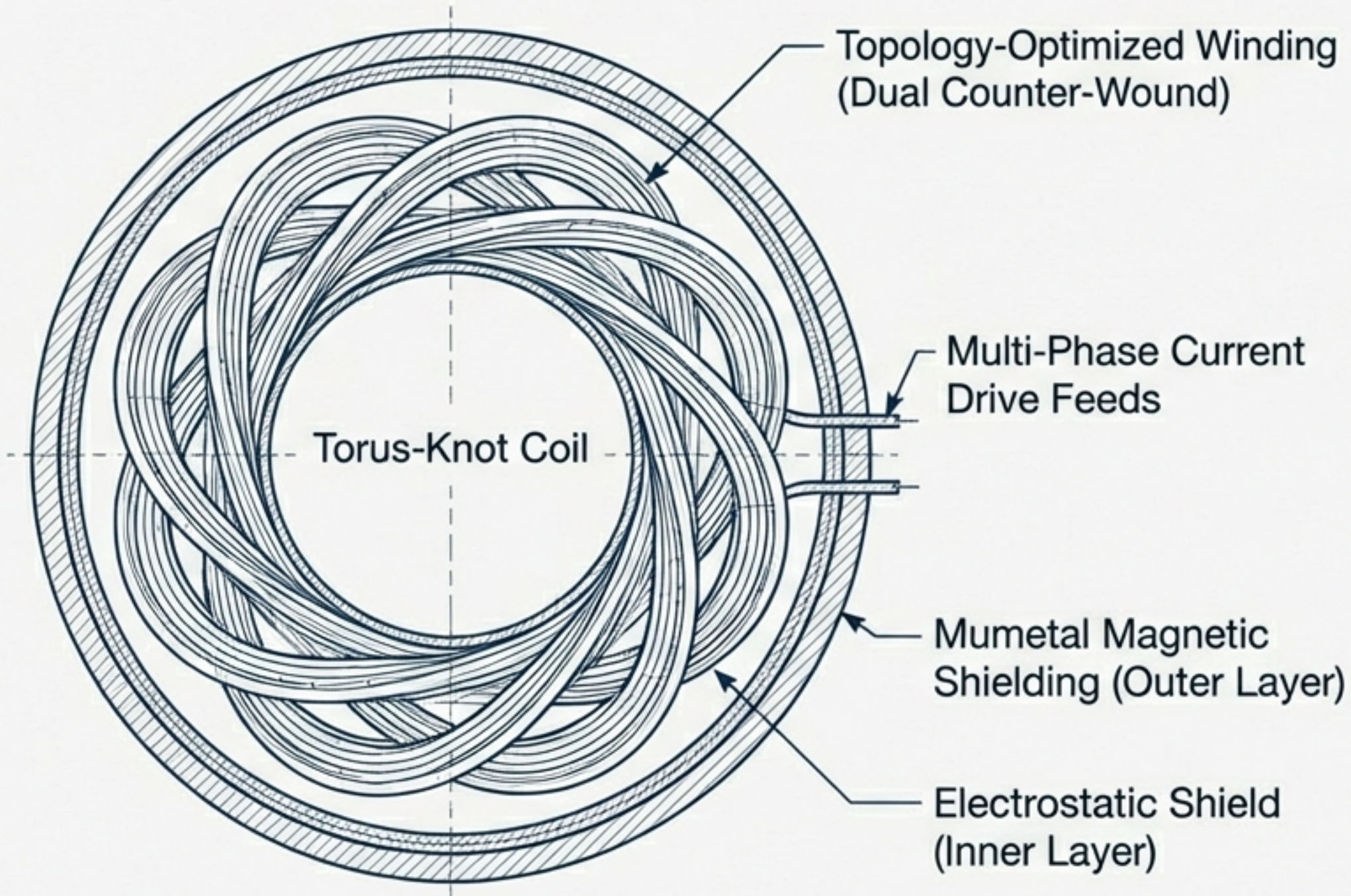
The perturbative nature ensures numerical stability without requiring full quantum-field simulations.

Order of Magnitude: A Precision Measurement Challenge fonenit Challenge

Parameter	Value	Context
Input Fields	$E \sim 10^7 \text{ V/m}, B \sim 10 \text{ T}$	Achievable in pulsed lab systems
Vacuum Density Change	$\Delta\rho/\rho \sim 10^{-36}$	Extremely weak QED coupling
Effective Acceleration	$ g_{\text{eff}} \lesssim 10^{-18} \text{ m/s}^2$	Comparable to state-of-the-art null-force limits

Takeaway: This is not a propulsion drive. It is a fundamental physics test at the edge of measurable sensitivity.

Experimental Apparatus: The Emitter

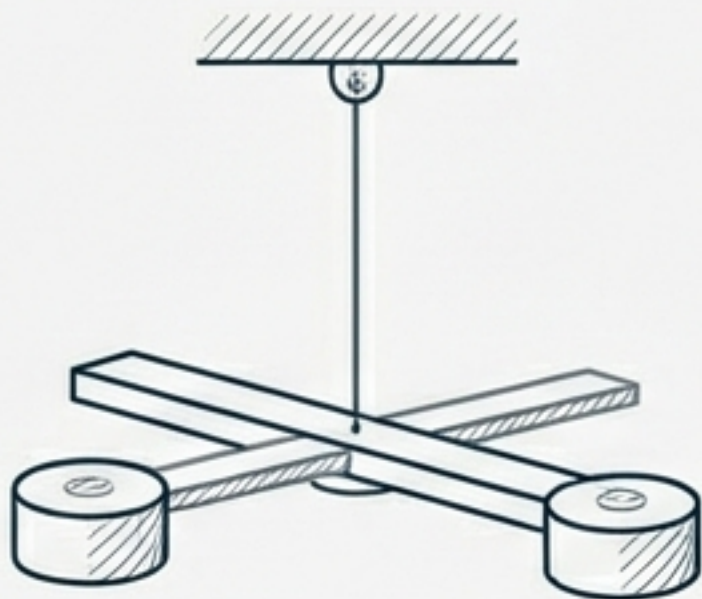


Design Features

1. **Anapole Configuration:**
Winding geometry cancels dipole moments.
2. **Dynamic Drive:**
Multi-phase excitation creates a rotating $T(t)$ moment.
3. **Noise Suppression:**
Heavy shielding prevents Lorentz force contamination.

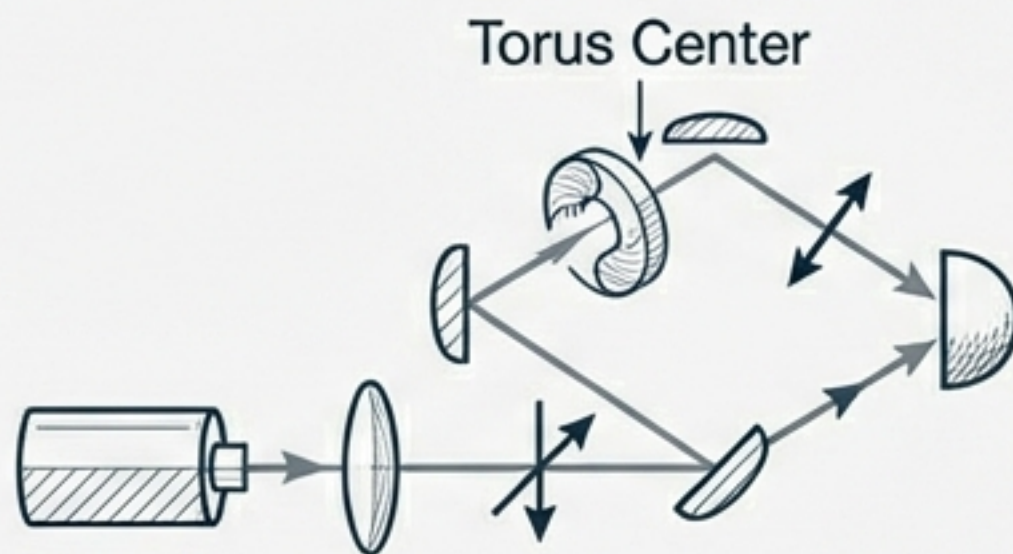
Measurement Channels: Detecting the Invisible

Force Balance



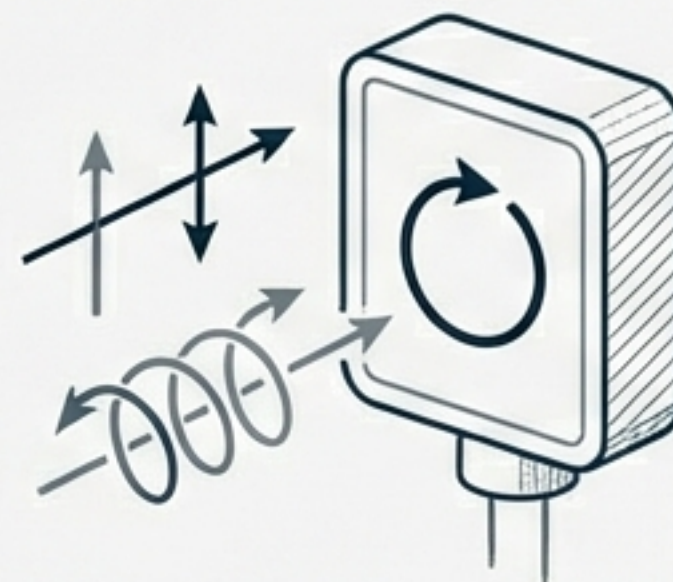
Search for phase-locked force signals at the drive frequency using a cantilever or torsion balance.

Vacuum Birefringence



Interferometric measurement of polarization rotation through the torus center (probing Δn).

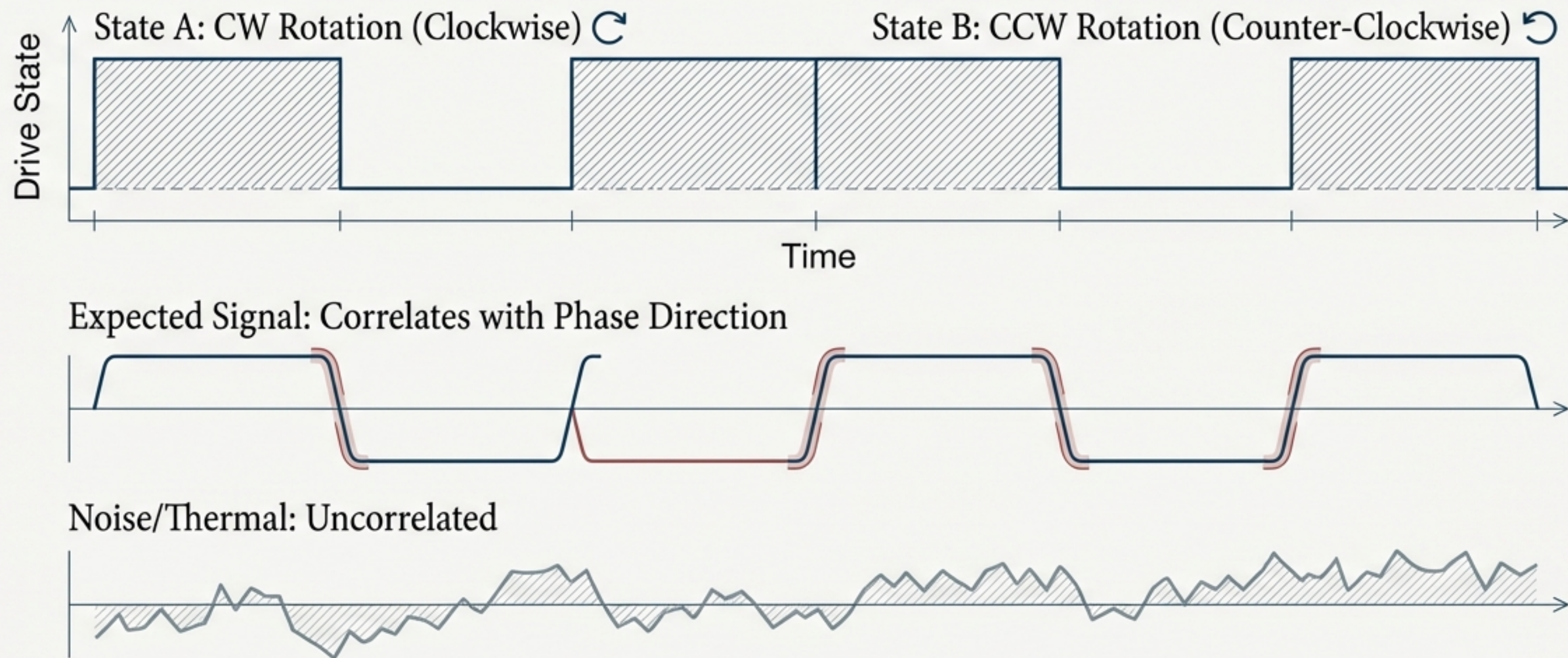
Parity-Sensitive Detection



Detecting weak circular/elliptical polarization components correlated with the drive phase.

The Critical Protocol: Differential Null-Tests

Differential Null-Tests



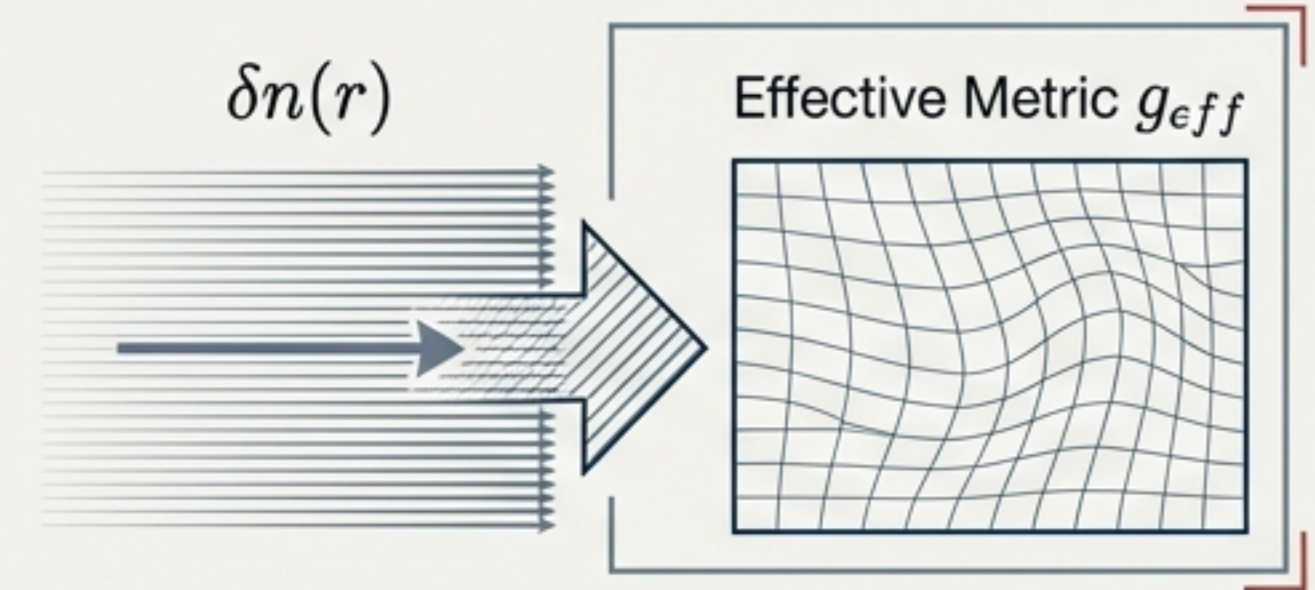
Validation: Measurements are taken under phase reversal. Only the differential signal matters. This rules out conventional thermal or mechanical noise.

Interpreting Results: Analog Gravity vs. Curved Spacetime

Inside the Polarizable Vacuum (PV) Formalism:

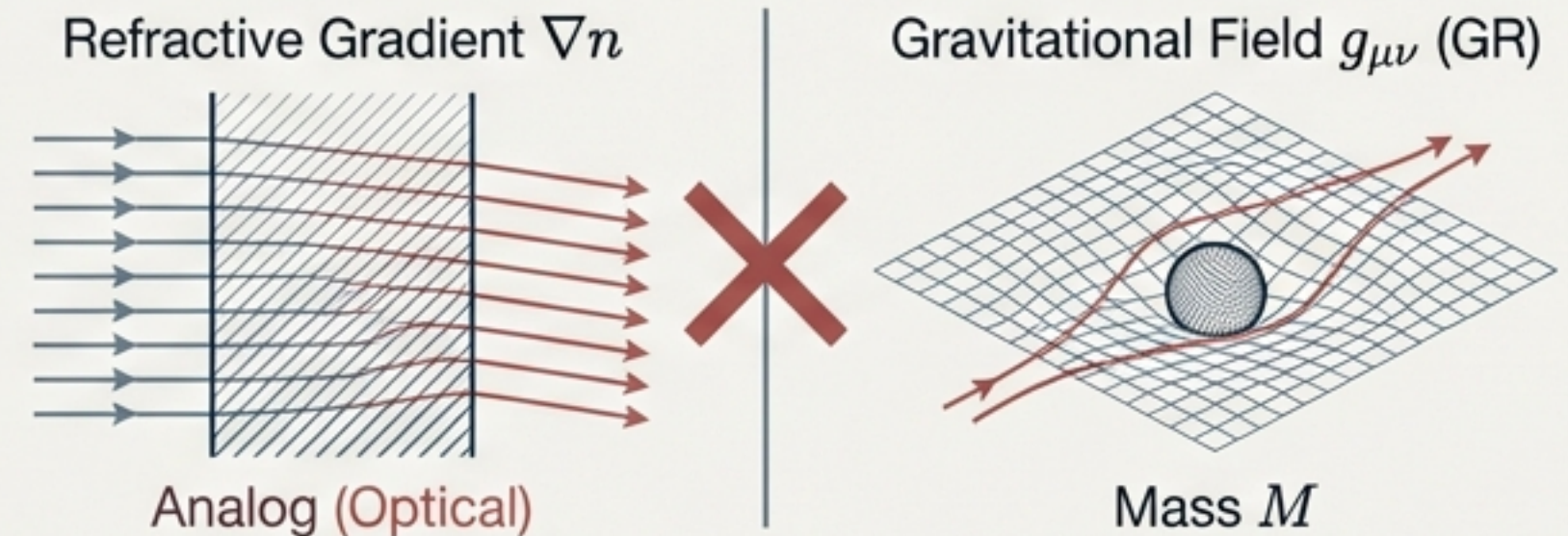
- Refractive Index Gradients $\equiv$ Effective Metric Perturbation

$$\delta n(r) \rightarrow g_{eff}$$



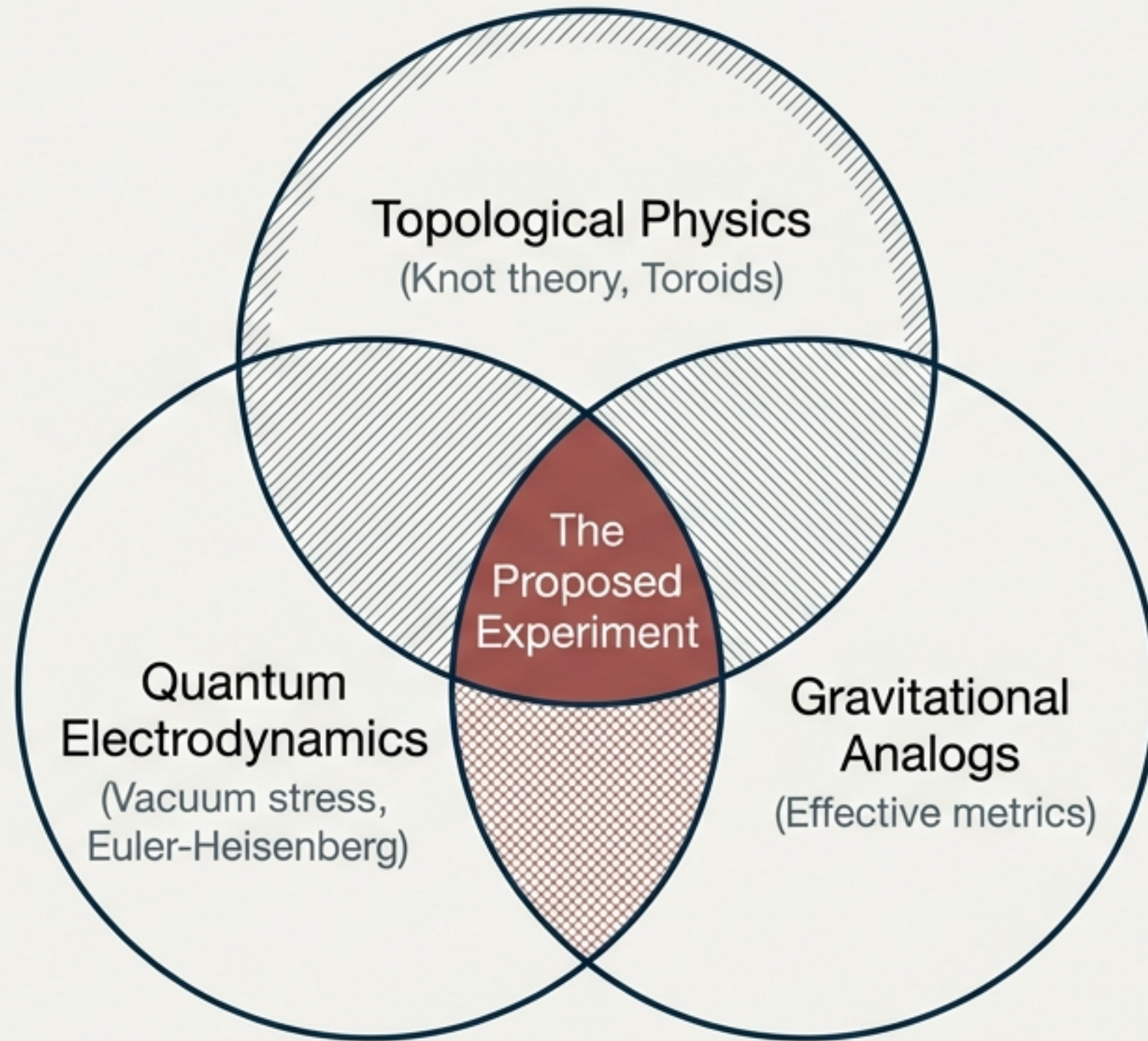
General Relativity Context:

Refractive-index gradients are *not* generally equivalent to gravitational fields outside of analog models.



A positive result demonstrates a coupling between Topological Electrodynamics and Vacuum Stress. It does not imply immediate macroscopic gravity control, but validates the interaction mechanism.

Why This Matters: A New Experimental Pathway



- **Falsifiability:** Moves vacuum engineering from theory to lab-bench.
- **New Physics:** Probes the specific coupling between topology and vacuum stress.
- **Foundational:** Tests Euler-Heisenberg predictions in limits previously inaccessible.

Conclusion: The Rigorous Frontier

Dynamic toroidal multipole fields offer a theoretically well-defined method for concentrating QED vacuum nonlinearities. While the predicted effect is extremely weak ($\sim 10^{-18}$ m/s²), this proposal creates a conservative, falsifiable framework.

We are proposing a first step toward understanding the fabric of spacetime through the lens of topological electrodynamics.

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Omar Iskandarani (2026)